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INTRODUCTION

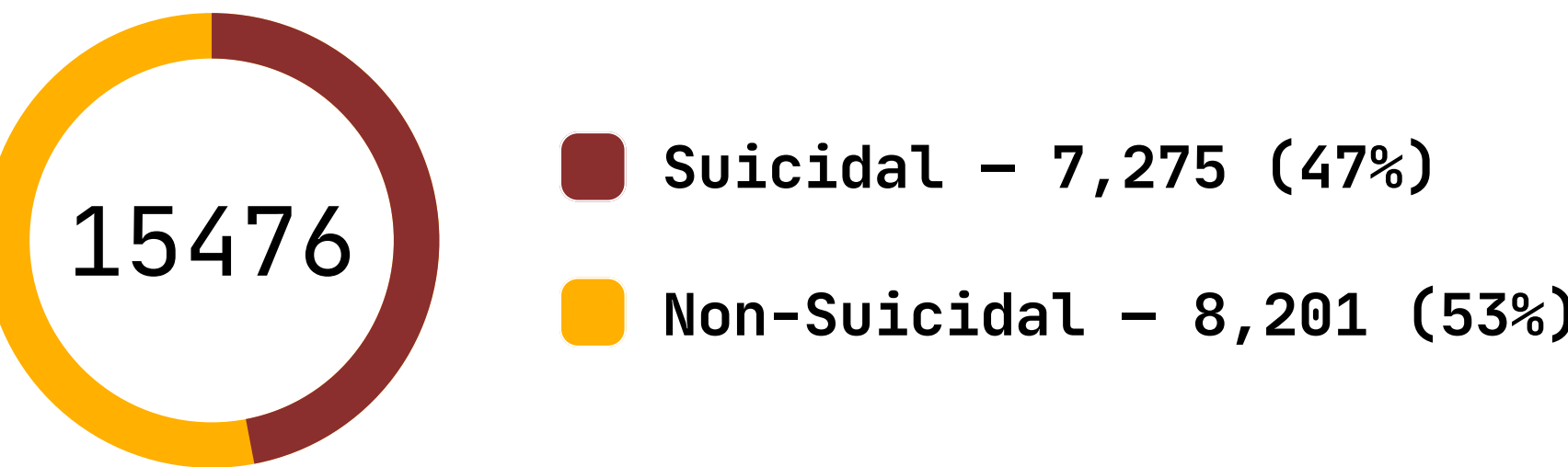
Suicide is one of the **leading causes of death** worldwide, yet it is often unseen or disregarded, and people who are in danger rarely reach out. Reddit has become a medium for individuals to vent their distress. This project uses these posts to assess whether machine learning models can learn to recognize these signals.

OBJECTIVE

This project applies four supervised machine learning models, **K-Nearest Neighbors**, **Logistic Regression**, **Naïve Bayes**, and **Neural Networks**, to classify **Reddit posts** as suicidal or non-suicidal. It evaluates their performance and the impact of key design choices and hyperparameters in detecting suicidal ideation from text.

DATASET

15,476 Reddit posts labelled Suicidal / Non-Suicidal. Sourced from suicidal subreddits (r/SuicideWatch, r/depression) and matched non-suicidal communities. Near-balanced split: 47% vs. 53%.



UNIQUE ASPECTS

- 1

NLP FEATURE ENGINEERING

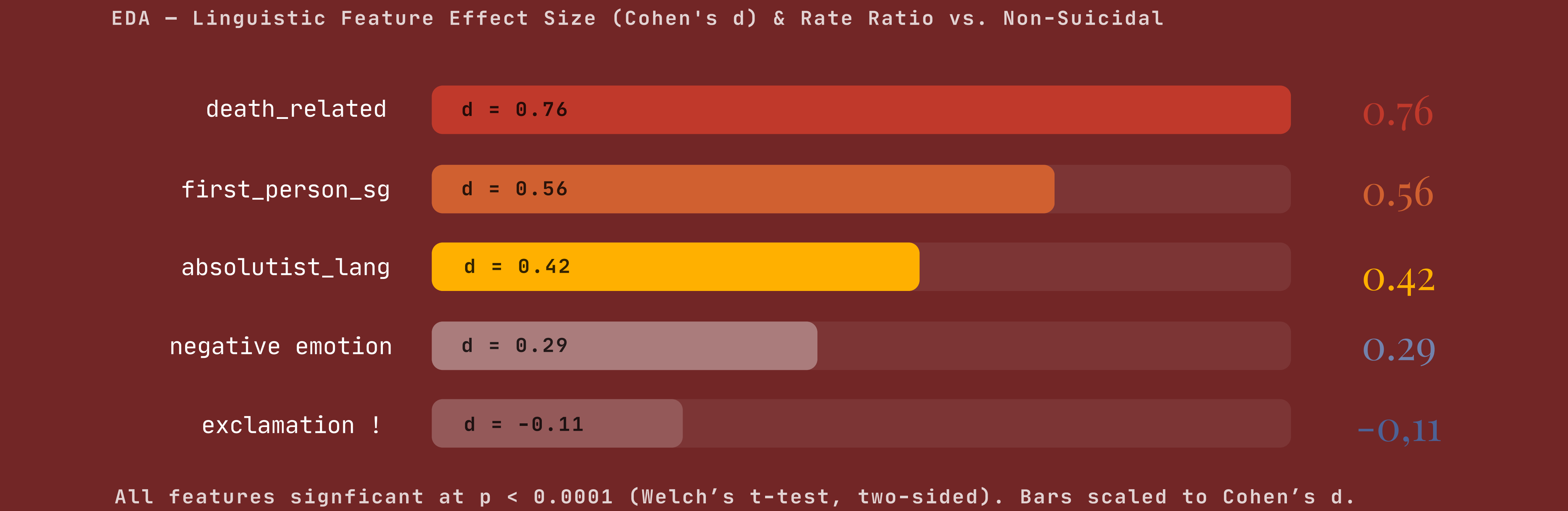
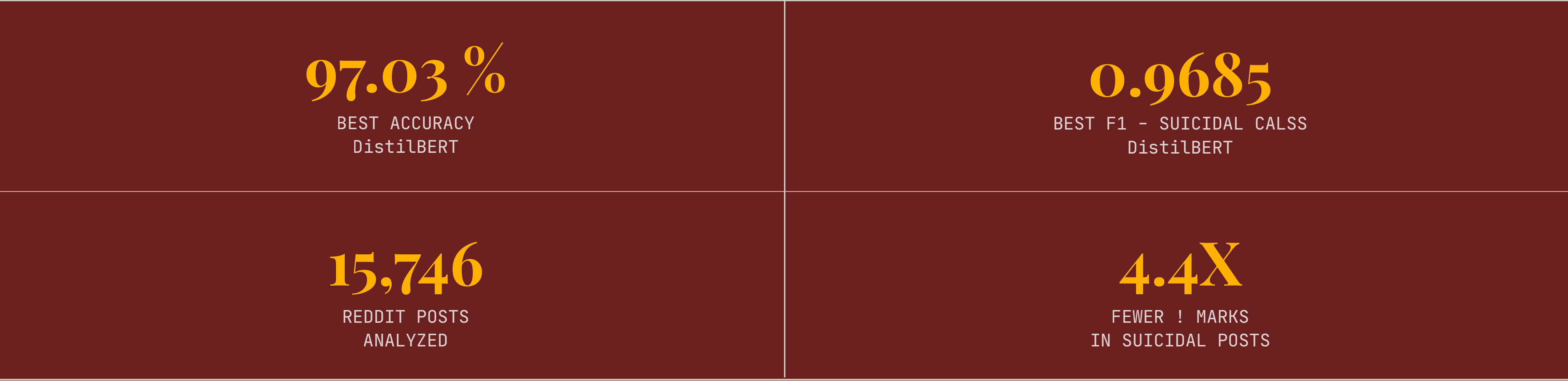
The project transformed raw text into features that capture **tone**, **self-focus**, and **emotional patterns** that signal suicidal language.
- 2

TWO-STAGE DEEP LEARNING

The project directly compared frozen **DistilBERT embeddings** fed into a tuned MLP against full end-to-end fine-tuning, isolating the specific contribution of task-adapted transformer weights to suicidality detection.

"Not every cry for help sounds like one."

Suicidal posts are not always explicit. While **death-related language proved to be the strongest single discriminator** between classes (Cohen's d = 0.76), at least 25% of suicidal posts contained zero death-related terms , expressing ideation instead through **euphemism**, **resignation**, and **implicit references** that no keyword list can reliably catch. **Traditional models trained on word frequency alone showed limitations:** Logistic Regression achieved 90.87% F1(S) but missed 176 suicidal posts, and even the strongest lexical model, **Naive Bayes with TF-IDF bigrams**, still missed 72. **Fine-tuned DistilBERT** reduced this to 39 missed posts by reading meaning rather than counting words. But, the gap that remained reflects a dataset where some posts are written, intentionally or not, to not sound like what they are. **Suicidality detection is not a vocabulary problem. It is a comprehension problem.**



BEST MODEL - FINE-TUNED DISTILBERT

Fine-tuned DistilBERT is the **best model**. It achieves the **highest F1** on the **Suicidal class**, the **highest macro F1**, and the **fewest false negatives**. The performance gain over frozen BERT + NN confirms that **end-to-end fine-tuning** adapting the transformer weights to **suicidality-specific language**, provides a meaningful improvement over using BERT purely as a static feature extractor.

RESULTS - MODEL COMPARISON

DistilBERT BEST			
ACCURACY	PRECISION	RECALL	F1(S)
97.03 %	0.9639	0.9732	0.9685

Naive Bayes			
ACCURACY	PRECISION	RECALL	F1(S)
94.74 %	0.9383	0.9505	0.9443

Logistic Regression			
ACCURACY	PRECISION	RECALL	F1(S)
91.70 %	0.9404	0.8790	0.9087

K - Nearest Neighbors			
ACCURACY	PRECISION	RECALL	F1(S)
84.53 %	0.8249	0.8515	0.8380

Recall = missed suicidal posts, precision = false alarms. Recall is the priority metric

KEY FINDINGS

- Fine-tuned DistilBERT is the best model**, achieving the highest F1 and fewest false negatives. Adapting transformer weights to the task is what drives improvement over frozen BERT.
- The **lexical ceiling is high**. Logistic Regression reaches ~93-95% F1 through TF-IDF alone, proving death-related language is powerful enough for linear models.
- False negatives are the hardest problem**. At least 25% of suicidal posts contain zero death-related terms, a floor no model fully overcomes.

CONCLUSION

Fine-tuned DistilBERT showed that context-aware understanding is the strongest approach for this task, but its improvement over simpler lexical models was modest. This suggests that suicidal language is often explicit enough for TF-IDF methods to capture effectively. The real challenge lies in cases that models tend to miss, such as implicit expressions of intent, irony, and posts without any death-related words. Addressing these cases will likely require richer training data and longer context rather than more complex model architectures.